

# PAPER\_962: Resonant Gravity Triadic Mode

**Author:** Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic\_solutions\_next.py (ResonantGravityTriadic) **Calculator:** ResonantGravityTriadicCalc (CP4 #546) **CVW:** v2.0.0 compliant

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## Abstract

The Resonant Gravity Triadic mode uses the 1.25 THz phonon linewidth  $\Gamma$  to tune neutron-drop and buoyancy reversal. When  $\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}}$ , the neutron drip-line shifts, controlling NS merger dynamics.

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## 1. Resonant Phonon Occupation

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}([SSq])$$

## 2. Neutron-Drop Threshold

$$\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}} \implies \text{neutron-drop triggered}$$

## 3. Buoyancy Reversal

$$t_{\text{rev}} = \frac{\pi}{2\Gamma}$$

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## References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
  2. PAPER\_961 — Compressed Gravity Triadic
  3. PAPER\_963 — Buoyancy Gravity Triadic
  4. PAPER\_966 — Unified Triadic Solver
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## Cross-Links

| Related Paper | Relationship                       |
|---------------|------------------------------------|
| PAPER_961     | Compressed branch of same triadic  |
| PAPER_963     | Buoyancy branch                    |
| PAPER_966     | Unified solver combining all three |
| PAPER_955     | Phonon resonance frequency link    |

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## Calibration Constants

| Constant             | Symbol | Value                                 | Validation Domain |
|----------------------|--------|---------------------------------------|-------------------|
| $\kappa$             | —      | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Damping           |
| $[SSq]$              | —      | 0.57                                  | String coupling   |
| $\beta_i$            | —      | 0.603                                 | Buoyancy          |
| $\Phi_{\text{crit}}$ | —      | Neutron-drop threshold                | Phase transition  |

## SM Anchor — CVW v2.0.0

| Observable                                     | UQFF Prediction                       | Status  |
|------------------------------------------------|---------------------------------------|---------|
| $g_{\text{res}}$ peak at $\omega_{\text{SCm}}$ | Resonance amplification               | Derived |
| $\Phi_{\text{crit}}$ neutron-drop              | Phase boundary for neutron star cores | Novel   |

## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

**Sector:** Resonant Gravity (5-Frequency Multi-Mode)

### §A.2 Lagrangian Density

$$\mathcal{L}_{\text{res}} = g_{\text{comp}}(r) \cdot \prod_{f \in \{\text{Super, Quantum, Aether, Fluid, Exp}\}} (1 + A_f \sin(\omega_f t + \phi_f))$$

### §A.3 Euler-Lagrange Equation of Motion

$$g_{\text{res}}(r, t) = g_{\text{comp}}(r) \prod_{f=1}^5 (1 + A_f \sin(\omega_f t + \phi_f)), \quad \Phi_{\text{crit}} : g_{\text{res}} > g_{\text{neutron-drop}}$$

### §A.4 Cosmogenesis Linkage Chain

PAPER\_877  $\rightarrow$  compressed gravity  $\rightarrow$  5-frequency modulation  $\rightarrow$  resonant amplification  $\rightarrow \Phi_{\text{crit}}$  phase boundary

## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

Resonance peaks modulate VDS by factor  $\prod (1 + A_f)$ .

### §B.2 DVP

Five frequencies correspond to five dipole vortex modes.

### §B.3 BSH

$g_{\text{res}}$  bounded above by  $g_{\text{comp}} \cdot \prod(1 + A_f)$  (constructive limit).

### §B.4 Production-Scale Consistency

| Metric        | Value         | Status    |
|---------------|---------------|-----------|
| 5 frequencies | All modulated | Confirmed |
| $[SSq]$       | 0.57          | Confirmed |